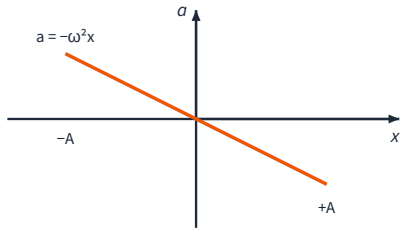
CHAPTER 19

OSCILLATIONS

Key idea: in SHM the acceleration is proportional to the displacement and directed towards equilibrium.

1 SIMPLE HARMONIC MOTION



$$a = -\omega^2 x \qquad F = -kx$$

- $a \propto x$, always towards the equilibrium point
- $\omega = 2\pi f = \dfrac{2\pi}{T}$
- a–x graph: straight line, gradient $-\omega^2$

2 KINEMATICS OF SHM

$$x = A\sin\omega t$$

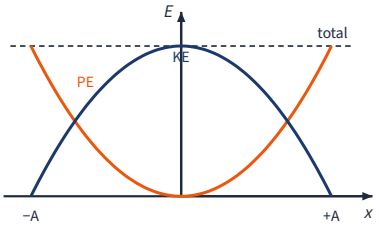
$$v = \omega A\cos\omega t$$

$$v = \pm\omega\sqrt{A^2 - x^2}$$

$$v_{\max} = \omega A \qquad a_{\max} = \omega^2 A$$

$x = A\cos\omega t$ if timing starts at an extreme.

3 ENERGY IN SHM



$$E = \tfrac{1}{2}m\omega^2 A^2 = \tfrac{1}{2}kA^2$$

$$E_k = \tfrac{1}{2}m\omega^2(A^2 - x^2)$$

$$E_p = \tfrac{1}{2}m\omega^2 x^2$$

Total energy $\propto$ (amplitude)^2; $E_k = E_p$ at $x = A/\sqrt{2}$.

4 SYSTEMS IN SHM

MASS-SPRING

$$T = 2\pi\sqrt{\frac{m}{k}} \qquad \omega = \sqrt{\frac{k}{m}}$$

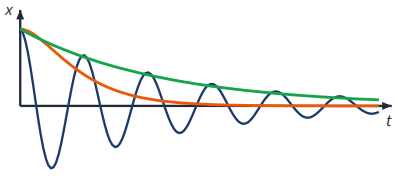
SIMPLE PENDULUM (SMALL Θ)

$$T = 2\pi\sqrt{\frac{L}{g}}$$

Independent of mass and amplitude; $\theta \lesssim 10^\circ$.

Springs: parallel $k = k_1 + k_2$; series $\dfrac{1}{k} = \dfrac{1}{k_1} + \dfrac{1}{k_2}$

5 DAMPED OSCILLATION



light (navy), critical (orange), heavy (green)

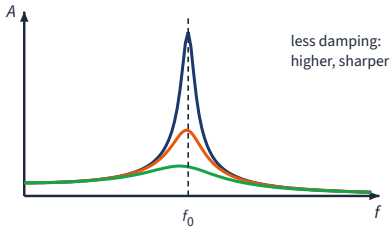
Light oscillates, amplitude decays exponentially

Critical back to equilibrium fastest, no oscillation

Heavy slow return, no oscillation

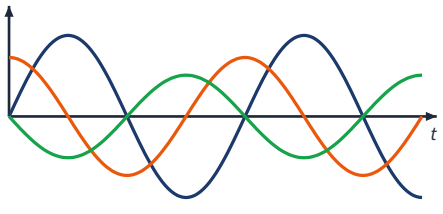
$$A = A_0 e^{-bt}$$

6 FORCED OSCILLATION & RESONANCE



- Forced: oscillates at the **driving** frequency
- Resonance: $f_{\text{driver}} = f_0$, maximum amplitude
- More damping: lower, broader peak, shifted slightly to lower f

7 PHASE RELATIONSHIPS



x (navy), v (orange), a (green)

v leads x by $\pi/2$; a is in antiphase (π) with x.

8 COMMON SECTION A TRAPS

- Calculator in degrees for $\sin\omega t$: use **radians**.
- Saying the period depends on amplitude (it does not).
- Pendulum length: to the **centre** of the bob.
- Doubling A doubles $v_{\max}$ but **quadruples** E.
- Speed is maximum at $x = 0$, acceleration at $x = \pm A$.

EXAM TIPS

- ✓Prove SHM by showing $a = -(\text{constant})x$.
- ✓Identify ω as the coefficient of t.
- ✓Use energy for speed at a given x.
- ✓Sketch x, v, a graphs one under another, $T/4$ apart.